

ICS Search Engine CS 121 Milestone 3

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Index Analytics

Metric	Value
Documents indexed	38,269
Unique tokens (with bigrams)	5,349,286
Index size	1,109,235 KB (~1.1 GB)
Exact duplicates skipped	3,771
Near-duplicates skipped (SimHash)	13,353
Partial index flushes	1,137

Query Results Summary

We evaluated 20 queries: 12 performed well, 8 performed poorly. Of the poor queries, 6 showed measurable improvement after code changes.

#	Query	Performance	Status
1	cristina lopes	Poor	Known limitation
2	machine learning	Good	Good from the start
3	ACM	Poor	Known limitation
4	master of software engineering	Good	Improved by code fix
5	data structures	Poor	Improved by code fix
6	artificial intelligence	Good	Good from the start
7	information retrieval	Good	Good from the start
8	neural networks	Good	Good from the start
9	graph algorithms	Good	Good from the start
10	web crawling	Poor	Improved by code fix
11	computer science phd	Poor	Known limitation
12	software engineering internship	Good	Improved by code fix
13	ics faculty	Poor	Known limitation
14	database systems	Good	Improved by code fix
15	operating systems	Good	Good from the start
16	compilers	Poor	Improved by code fix
17	python programming	Good	Good from the start

18	undergraduate admissions	Good	Good from the start
19	graduate programs	Good	Good from the start
20	cyber security	Poor	Known limitation

Top 10 Results for Each Query

Query 1: "cristina lopes" [Poor]

#	URL
1	https://www.ics.uci.edu/community/news/notes/notes_2010.php
2	https://www.ics.uci.edu/faculty/area/index.php
3	https://www.ics.uci.edu/community/news/notes/notes_2009.php
4	https://www.ics.uci.edu/community/news/notes/notes_2011.php
5	https://www.ics.uci.edu/community/news/view_news.php
6	https://www.ics.uci.edu/~taylor/classes/211/syllabusFQ06.html
7	https://www.ics.uci.edu/~taylor/classes/211/syllabusFQ07.html
8	https://www.ics.uci.edu/community/news/spotlight/spotlight_microsoftspellerchallenge.php
9	https://www.ics.uci.edu/~hsajnani/
10	https://www.informatics.uci.edu/explore/facts-figures/

Query 2: "machine learning" [Good]

#	URL
1	https://www.ics.uci.edu/~pazzani/Publications/OldPublications.html
2	https://www.ics.uci.edu/~pazzani/Publications/APubs.html
3	https://www.ics.uci.edu/community/news/view_news.php
4	https://www.ics.uci.edu/~welling/publications/publications.html
5	https://www.ics.uci.edu/~smyth/papers.html
6	https://www.ics.uci.edu/~rickl/courses/cs-171/2016-fq-cs171/CS-171-FQ-2016_draft.htm
7	https://www.ics.uci.edu/~rickl/courses/cs-171/2014-fq-cs171/CS-171-FQ-2014.htm
8	https://www.ics.uci.edu/~pazzani/RTF/AAAI.html
9	https://www.ics.uci.edu/~smyth/courses/cs274/notes.html
10	https://www.ics.uci.edu/~pazzani/Research.html

Query 3: "ACM" [Poor]

#	URL
1	https://www.ics.uci.edu/~eppstein/pubs/j-no.html
2	https://www.ics.uci.edu/~eppstein/pubs/c-no.html
3	https://www.ics.uci.edu/~eppstein/pubs/j-talg.html

4	https://www.ics.uci.edu/~eppstein/pubs/year.html
5	https://www.ics.uci.edu/~eppstein/pubs/future.html
6	https://www.ics.uci.edu/~eppstein/pubs/all.html
7	https://www.ics.uci.edu/~kay/courses/131/s04readings.html
8	https://www.ics.uci.edu/~gmark/Home_page/Publications.html
9	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html
10	https://www.ics.uci.edu/~eppstein/pubs/geom-all.html

Query 4: "master of software engineering" [Good]

#	URL
1	https://www.ics.uci.edu/grad/admissions/Prospective_ApplicationProcess.php
2	https://www.ics.uci.edu/~nenovita/vita.html
3	https://www.informatics.uci.edu/grad/ms-software-engineering/
4	https://www.ics.uci.edu/~taylor/Publications.htm
5	https://www.ics.uci.edu/~pattis/quotations.html
6	https://www.ics.uci.edu/~andre/publications.html
7	https://www.ics.uci.edu/~seal/publications.html
8	https://www.ics.uci.edu/~emilyo/publications.html
9	https://www.ics.uci.edu/~wscacchi/Papers/Vintage/Software_Productivity.html
10	https://www.ics.uci.edu/~wscacchi/publications.html

Query 5: "data structures" [Poor]

#	URL
1	https://www.ics.uci.edu/~eppstein/pubs/all.html
2	https://www.ics.uci.edu/~eppstein/pubs/j-no.html
3	https://www.ics.uci.edu/~eppstein/pubs/geom-all.html
4	https://www.ics.uci.edu/~eppstein/pubs/graph-all.html
5	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/api/annotated.html
6	https://www.ics.uci.edu/~eppstein/pubs/c-wads.html
7	https://www.ics.uci.edu/~goodrich/pubs/index.html
8	https://www.ics.uci.edu/~eppstein/pubs/a-goodrich.html
9	https://www.ics.uci.edu/~eppstein/pubs/a-solo.html
10	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html

Query 6: "artificial intelligence" [Good]

#	URL
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1	https://www.ics.uci.edu/~dechter/publications.html
2	https://www.ics.uci.edu/~pazzani/Publications/OldPublications.html
3	https://www.ics.uci.edu/~pazzani/Publications/APubs.html
4	https://www.ics.uci.edu/~ihler/papers/bib.html
5	https://www.ics.uci.edu/~rickl/courses/cs-171/2014-wq-cs171/CS-171-WQ-2014.htm
6	https://www.ics.uci.edu/~rickl/courses/cs-171/2013-fq-cs171/CS-171-FQ-2013.htm
7	https://www.ics.uci.edu/~ihler/pubs00.html
8	https://www.ics.uci.edu/~rickl/courses/cs-171/2015-fq-cs171/CS-171-FQ-2015.htm
9	https://www.ics.uci.edu/~rickl/courses/cs-171/2016-wq-cs171/CS-171-WQ-2016.htm
10	https://www.ics.uci.edu/~rickl/courses/cs-171/2014-fq-cs171/CS-171-FQ-2014.htm

Query 7: "information retrieval" [Good]

#	URL
1	https://www.ics.uci.edu/~gbowker/converge.html
2	https://www.ics.uci.edu/~pazzani/RTF/AAAI.html
3	https://www.ics.uci.edu/~pazzani/Syskill.html
4	http://sprout.ics.uci.edu/projects/iarpa-app/related.html
5	https://www.ics.uci.edu/~welling/NSFcareer/NSFcareer.html
6	https://www.ics.uci.edu/~kay/courses/i141/refs.html
7	https://www.ics.uci.edu/~rohit/IEEE-L7-http-gopher.html
8	https://www.ics.uci.edu/~welling/publications/publications.html
9	https://www.ics.uci.edu/~sharad/projects.html
10	https://www.ics.uci.edu/ugrad/degrees/degree_cs.php

Query 8: "neural networks" [Good]

#	URL
1	https://www.ics.uci.edu/~emj/Mjolsness_CV_V67p.htm
2	https://www.ics.uci.edu/~welling/publications/publications.html
3	https://www.ics.uci.edu/~rickl/rickl-patent-5526281.html
4	https://www.ics.uci.edu/~ihler/papers/abs.html
5	https://www.ics.uci.edu/~xhx/publications.html
6	https://www.ics.uci.edu/~pazzani/CogSci95.html
7	https://www.ics.uci.edu/~dchudova/research.htm
8	https://www.ics.uci.edu/~pjsadows/research.html
9	https://www.ics.uci.edu/~theory/269/190524b.html
10	https://www.ics.uci.edu/~ihler/pubs00.html

Query 9: "graph algorithms" [Good]

#	URL
1	https://www.ics.uci.edu/~eppstein/pubs/all.html
2	https://www.ics.uci.edu/~eppstein/pubs/graph-all.html
3	https://www.ics.uci.edu/~eppstein/pubs/c-no.html
4	https://www.ics.uci.edu/~eppstein/pubs/geom-all.html
5	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html
6	https://www.ics.uci.edu/~eppstein/pubs/j-jga.html
7	https://www.ics.uci.edu/~eppstein/pubs/gdraw.html
8	https://www.ics.uci.edu/~eppstein/pubs/a-solo.html
9	https://www.ics.uci.edu/~eppstein/pubs/graph-planar.html
10	https://www.ics.uci.edu/~eppstein/pubs/c-gd.html

Query 10: "web crawling" [Poor]

#	URL
1	https://www.ics.uci.edu/~kay/courses/i141/w15.html
2	https://grape.ics.uci.edu/wiki/public/wiki/cs221-2019-spring
3	https://www.ics.uci.edu/~ics1c/doc/spiders.html
4	https://www.ics.uci.edu/~kay/courses/i141/hw/asst3.html
5	https://www.ics.uci.edu/~pattis/ICS-21/handouts/syllabus/syllabus.html
6	https://www.ics.uci.edu/community/news/notes/notes_2010.php
7	https://www.ics.uci.edu/~pattis/ICS-46/handouts/syllabus/syllabus.html
8	https://www.ics.uci.edu/~pattis/ICS-33/handouts/syllabus/syllabus.html
9	https://www.ics.uci.edu/~pattis/ICS-31/handouts/syllabus/syllabus.html
10	http://vision.ics.uci.edu/sccv/2009/schedule.html

Query 11: "computer science phd" [Poor]

#	URL
1	https://cml.ics.uci.edu/category/news/page/6/
2	https://cml.ics.uci.edu/tag/news/page/4/
3	https://cml.ics.uci.edu/2009/04/2009_fellowships/
4	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html
5	https://www.ics.uci.edu/~khodabah/
6	https://www.ics.uci.edu/community/news/notes/notes_2010.php
7	https://www.ics.uci.edu/community/news/notes/notes_2007.php

8	https://www.ics.uci.edu/community/news/notes/notes_2009.php
9	https://www.ics.uci.edu/~stasio/
10	https://www.ics.uci.edu/~smyth/research_group.html

Query 12: "software engineering internship" [Good]

#	URL
1	https://www.ics.uci.edu/~taylor/Publications.htm
2	https://www.ics.uci.edu/~andre/publications.html
3	https://www.ics.uci.edu/~emilyo/publications.html
4	https://www.ics.uci.edu/~seal/publications.html
5	https://www.ics.uci.edu/~emilyo/SimSE/publications.html
6	https://www.ics.uci.edu/~wscacchi/publications.html
7	https://www.ics.uci.edu/~fielding/pubs/dissertation/references.htm
8	https://www.ics.uci.edu/ugrad/degrees/degree_se.php
9	https://www.ics.uci.edu/~wscacchi/Papers/Vintage/Software_Productivity.html
10	https://www.ics.uci.edu/community/news/notes/notes_2007.php

Query 13: "ics faculty" [Poor]

#	URL
1	https://www.ics.uci.edu/~rickl/courses/ics-h197/2016-fq-h197/ICS-H197-FQ-2016.htm
2	https://www.ics.uci.edu/~rickl/courses/ics-h197/2013-fq-h197/ICS-H197-FQ-2013.htm
3	https://www.ics.uci.edu/~rickl/courses/ics-h197/2011-fq-h197/ICS-H197-FQ-2011.htm
4	https://www.ics.uci.edu/~rickl/courses/ics-h197/2014-fq-h197/ICS-H197-FQ-2014.htm
5	https://www.ics.uci.edu/community/news/view_news.php
6	https://www.ics.uci.edu/community/news/notes/notes_2013.php
7	https://www.ics.uci.edu/~rickl/courses/ics-h197/syl.htm
8	https://www.ics.uci.edu/community/news/notes/notes_2014.php
9	https://www.ics.uci.edu/faculty/area/index.php
10	https://www.ics.uci.edu/~rickl/courses/ics-h197/syllabus.html

Query 14: "database systems" [Good]

#	URL
1	https://www.ics.uci.edu/~mjcarey/MJCarey_Publications.html
2	https://www.ics.uci.edu/~mjcarey/MJCarey_Invited_Lectures.html
3	https://www.ics.uci.edu/~mjcarey/MJCarey_PhD_Students_Supervised.html
4	https://www.ics.uci.edu/~wscacchi/Papers/Internet-Procurement/Internet_Procurement.html

5	https://www.ics.uci.edu/~dvk/pub.html
6	https://www.ics.uci.edu/~mjcarey/MJCarey_Professional_Activities.html
7	http://sprout.ics.uci.edu/past_projects/odb/index.html
8	https://www.ics.uci.edu/community/news/notes/notes_2009.php
9	https://www.ics.uci.edu/~ics184/index.html
10	http://sherlock.ics.uci.edu/qgd.html

Query 15: "operating systems" [Good]

#	URL
1	https://www.ics.uci.edu/~ics1c/hw4/31.html
2	https://www.ics.uci.edu/~aburtsev/index.html
3	https://www.ics.uci.edu/~cs237/reading/Millennium.htm
4	https://www.ics.uci.edu/~aburtsev/publications.html
5	https://www.ics.uci.edu/~aburtsev/238P/index.html
6	https://www.ics.uci.edu/~aburtsev/143A/2018fall/index.html
7	https://www.ics.uci.edu/~aburtsev/238P/2018fall/index.html
8	https://www.ics.uci.edu/~aburtsev/238P/2018winter/index.html
9	https://www.ics.uci.edu/~aburtsev/143A/2017winter/index.html
10	https://www.ics.uci.edu/~wscacchi/Papers/VISTA/VISTA.html

Query 16: "compilers" [Poor]

#	URL
1	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/compiling-unix.html
2	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/compiling.html
3	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/compiling-windows.html
4	https://www.ics.uci.edu/~dechter/courses/ics-275a/fall-99/slides/node67.html
5	https://www.ics.uci.edu/~dechter/courses/ics-275a/fall-99/slides/node63.html
6	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/api/host_8h.html
7	https://www.ics.uci.edu/~dechter/courses/ics-275a/fall-99/slides/node66.html
8	https://www.ics.uci.edu/~majumder/VC/211HW3/vlfeat/doc/api/portability.html
9	https://www.ics.uci.edu/~thornton/ics142/MidtermStudyGuide.html
10	https://www.ics.uci.edu/~dechter/courses/ics-275a/fall-99/slides/node49.html

Query 17: "python programming" [Good]

#	URL
1	https://www.ics.uci.edu/~pattis/ICS-33/lectures/overview/lecture.html

2	https://www.ics.uci.edu/~pattis/common/handouts/introtopythonineclipse/lecture.html
3	https://www.ics.uci.edu/~pattis/ICS-31/handouts/syllabus/syllabus.html
4	https://www.ics.uci.edu/~pattis/ICS-31/lectures/overview/lecture.html
5	https://www.ics.uci.edu/~pattis/ICS-33/handouts/syllabus/syllabus.html
6	https://www.ics.uci.edu/~thornton/ics32/ProjectGuide/Project3/index.html
7	https://www.ics.uci.edu/~pattis/ICS-33/assignments/program0/program.html
8	https://www.ics.uci.edu/~pattis/ICS-31/announcements.html
9	https://www.ics.uci.edu/~pattis/common/online33.html
10	https://www.ics.uci.edu/~pattis/ICS-33/handouts.html

Query 18: "undergraduate admissions" [Good]

#	URL
1	https://www.cs.uci.edu/undergraduate-programs/
2	https://www.informatics.uci.edu/very-top-footer-menu-items/site-map/
3	https://www.cs.uci.edu/ics-staff-faculty-honored-at-inaugural-faculty-staff-awards-celebration/
4	https://www.cs.uci.edu/%EF%BB%BFsingh-thornton-recognized-for-excellence-in-undergraduate-teaching/
5	https://www.cs.uci.edu/news/page/4/
6	https://www.cs.uci.edu/senior-spotlight-kevin-kes-journey-from-guangzhou-to-uci-to-amazon/
7	https://www.ics.uci.edu/~redmiles/ics125-FQ01/adp.html
8	https://www.cs.uci.edu/2017-ics-deans-award-winners/
9	https://www.cs.uci.edu/new-faculty-spotlight-professor-jennifer-wong-ma-shares-her-passion-for-teaching/
10	https://www.cs.uci.edu/faculty/

Query 19: "graduate programs" [Good]

#	URL
1	https://www.ics.uci.edu/community/news/notes/notes_2007.php
2	https://www.ics.uci.edu/community/news/view_news.php
3	https://www.ics.uci.edu/community/news/notes/notes_2014.php
4	https://www.ics.uci.edu/about/about_factsfigures.php
5	https://www.ics.uci.edu/~ics1c/hw3/03.html
6	https://www.ics.uci.edu/community/news/notes/notes_2011.php
7	https://www.ics.uci.edu/community/news/notes/notes_2012.php
8	https://www.ics.uci.edu/grad/policies/GradPolicies_GradStudentReview.php
9	https://www.ics.uci.edu/ugrad/QA_After_Graduation.php
10	https://www.ics.uci.edu/grad/financial/index.php

Query 20: "cyber security" [Poor]

#	URL
1	https://www.ics.uci.edu/community/events/openhouse/research12.html
2	https://www.ics.uci.edu/community/news/notes/notes_2012.php
3	https://www.ics.uci.edu/~gts/pubs.html
4	https://www.ics.uci.edu/~keldefra/pubs.html
5	https://www.ics.uci.edu/community/events/openhouse/demos.html
6	https://www.ics.uci.edu/community/news/notes/notes_2007.php
7	https://studentcouncil.ics.uci.edu/clubs.html
8	https://www.ics.uci.edu/~shantas/publications.html
9	http://sprout.ics.uci.edu/people/tkaczmar/index.html
10	https://www.ics.uci.edu/~kobsa/phds/phds.htm

Analysis of Poorly Performing Queries

cristina lopes : Known Limitation

The actual faculty profile page does not appear in the top 10. News archive pages dominate due to high link frequency. Anchor text indexing (ANCHOR_WEIGHT=1.5) and HITS authority scoring helped surface more Lopes-related pages, but named-entity boosting was not implemented to remain general.

ACM : Known Limitation

Results are dominated by Eppstein's publication list pages which frequently mention ACM journal names. ACM is an inherently ambiguous single-word query that TF-IDF alone cannot disambiguate by intent. Anchor text indexing helped surface a more diverse set of ACM-related pages.

data structures : Partially Fixed

Previously maven-contents.txt ranked #1. Removed by the file extension penalty (0.1x multiplier for .txt files). Bigram indexing for 'data_structur' now boosts pages containing the exact phrase. Eppstein's algorithm publication pages now dominate which is more relevant than raw dataset files.

web crawling : Fixed

Previously showed the same wiki URL repeated 8 times (version=107, 121, 122, etc.). Fixed by: (1) URL normalization strips query parameters so all wiki revision URLs collapse to one canonical URL, and (2) get_top_urls() deduplicates URLs in results. Results are now 10 distinct pages.

computer science phd : Known Limitation

The official CS PhD admissions page does not appear in the top 10. CML news category/tag pages dominate due to frequently mentioning PhD and computer science terms. HITS authority scoring slightly improved this by boosting authoritative department pages.

ics faculty : Known Limitation

Multiple year-versioned ICS honors course syllabi dominate. The faculty directory appears at #9. Year-versioned pages evade SimHash because 2011/2013/2014/2016 content variations exceed the Hamming threshold.

compilers : Partially Fixed

Previously maven-contents.txt ranked #1. Now removed by the file penalty. VLFeat build docs still dominate because Porter stemmer collapses 'compilers' and 'compiling' to the same root. ICS 142 (compiler course) now appears at #9.

cyber security : Known Limitation

Results are individual faculty publication pages mentioning security. UCI does not have a standalone cybersecurity department so the corpus genuinely lacks an authoritative page for this query. Anchor text indexing and HITS authority helped surface security-focused pages.

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